

```
C:\Users\minfa\AI Test\Write-ups\TokenAwakening>python eval_prompts.py
C:\Users\minfa\AI Test\Write-ups\TokenAwakening\eval_prompts.py:11:
DeprecationWarning: 'LLMTestCaseParams' is deprecated and will be removed
in a future release. Use 'SingleTurnParams' instead.
    from deepeval.test_case import LLMTestCase, LLMTestCaseParams
Extracting text from
C:\Users\minfa\Downloads\2025_Federal_Reserve_Stress_Test_Results.pdf...
Extracting text from
C:\Users\minfa\Downloads\2026_Federal_Reserve_Stress_Test_Results.pdf...
Extracting text from
C:\Users\minfa\Downloads\2025_Federal_Reserve_Stress_Test_Scenario.pdf...
Extracting text from
C:\Users\minfa\Downloads\2026_Federal_Reserve_Stress_Test_Scenario.pdf...
Extraction complete. Read 503586 characters.
```

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🏃 RUNNING COMPLEX BRIEF PROMPT | Model: gemini-2.5-flash
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[Output]

****Executive Summary: Analysis of Citigroup's Projected Loan Loss Rate Changes (2026 vs. 2025)****

As Chief Risk Officer, I have reviewed the 2025 and 2026 Federal Reserve Stress Test Results and Scenarios to understand the year-over-year variance in Citigroup's projected loan loss rates for its 'Credit cards' and 'Commercial real estate, domestic' (CRE) portfolios under the severely adverse scenario. The analysis attributes these changes to specific shifts in macroeconomic conditions outlined in the respective stress test scenarios.

****Citigroup's Projected Loan Loss Rates:****

*** **Credit Cards:****

* In the 2025 stress test, Citigroup's projected loan loss rate for credit cards was ****16.6 percent**** ("2025 CCAR RESULTS", Table A.8).

* In the 2026 stress test, this rate decreased slightly to ****16.3 percent**** ("2026 CCAR RESULTS", Table A.9).

*** **Commercial Real Estate (CRE), Domestic:****

* In the 2025 stress test, Citigroup's projected loan loss rate for domestic CRE was ****8.3 percent**** ("2025 CCAR RESULTS", Table A.8).

* In the 2026 stress test, this rate increased to ****10.9 percent**** ("2026 CCAR RESULTS", Table A.9).

****Attribution of Variance to Macroeconomic Changes:****

The observed changes in Citigroup's projected loan loss rates are directly attributable to the differing macroeconomic severities in the 2025 and 2026 severely adverse scenarios:

1. ****Credit Card Loan Loss Rate (Decrease from 16.6% to 16.3%):****

The slight decrease in the projected credit card loan loss rate for Citigroup is primarily driven by a ****less severe increase in the**

unemployment rate** in the 2026 severely adverse scenario compared to 2025.

* The 2025 severely adverse scenario featured an unemployment rate increase of **+5.9 percentage points** to a peak of 10% ("2025 CCAR RESULTS", Table 2).

* The 2026 severely adverse scenario featured a smaller unemployment rate increase of **+5.5 percentage points** to a peak of 10% ("2026 CCAR RESULTS", Table 2).

As credit card losses are highly sensitive to unemployment, a less pronounced rise in joblessness in the 2026 scenario would naturally lead to a marginally lower projected loss rate for this portfolio.

2. **Commercial Real Estate (CRE) Domestic Loan Loss Rate (Increase from 8.3% to 10.9%):**

The notable increase in Citigroup's projected domestic CRE loan loss rate is a direct consequence of a **significantly more severe decline in commercial real estate prices** and **heightened stress in corporate debt markets** in the 2026 severely adverse scenario.

* The 2025 severely adverse scenario projected a **↓30% decline in CRE prices** and a **↑3.9 percentage point increase in the BBB-bond rate spread** to 5.0% ("2025 CCAR RESULTS", Table 2).

* In contrast, the 2026 severely adverse scenario projected a more severe **↓39% decline in CRE prices** and a larger **↑4.7 percentage point increase in the BBB-bond rate spread** to 5.7% ("2026 CCAR RESULTS", Table 2).

The more substantial drop in CRE prices directly impacts the collateral value of CRE loans, increasing potential losses. Concurrently, the wider BBB-bond rate spread indicates greater stress in corporate debt markets, which can signal deteriorating conditions for commercial borrowers and their ability to service CRE debt, thus contributing to higher projected losses.

In summary, while Citigroup's credit card portfolio benefited from a slightly less severe unemployment shock in 2026, its domestic CRE portfolio faced a more challenging outlook due to a deeper projected decline in commercial real estate values and increased corporate credit market stress.

[Usage] Prompt Tokens: 168,376 | Output Tokens: 2,731

[Cost] Input Cost: \$0.050513 | Output Cost: \$0.006828 | Total Cost: \$0.057340

→ Macro numbers found. Running DeepEval Economic Reasoning Check...

→ DeepEval Score: 90.0% | Reason: The response strongly aligns with the evaluation steps. It accurately references YoY loss rate changes for both portfolios (Credit Cards decreasing from 16.6% to 16.3%, and CRE increasing from 8.3% to 10.9%), correctly preserving signs. It logically connects macroeconomic drivers to loss outcomes using sound reasoning: unemployment for credit cards (a less severe increase, +5.9pp vs +5.5pp, explaining the slight loss decrease) and CRE price decline plus BBB-bond spread for CRE (more severe -30% to -39% and wider spreads, explaining the loss increase). The direction and magnitude of changes are consistent with the stated severities, and causal relationships are clearly explained. A minor shortcoming is that the executive summary cites 'CCAR RESULTS' Table 2 for scenario variables rather than the 'Scenarios'

documents specified in the input, indicating a slight sourcing inconsistency, but the substantive analysis remains coherent and well-supported.

→ DeepEval Usage: Input 3,024 / Output 629 | Total Cost: \$0.030845

✅ Validation: PASSED

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🏃 RUNNING COMPLEX_BRIEF PROMPT | Model: gemini-2.5-pro

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[Output]

TO: Executive Management

FROM: Chief Risk Officer

DATE: July 1, 2026

SUBJECT: Executive Summary: Year-over-Year Attribution of Change in Projected Loan Loss Rates for Citigroup Credit Card and Commercial Real Estate Portfolios (2026 vs. 2025 CCAR)

This summary provides an attribution analysis for the year-over-year changes in Citigroup's projected loan loss rates for the Credit Card and Commercial Real Estate (CRE) portfolios under the Federal Reserve's Severely Adverse scenarios for 2026 and 2025. The variance in loss rates is directly attributable to specific changes in the macroeconomic assumptions outlined in the respective scenario documents.

Credit Cards Portfolio

Citigroup's projected loan loss rate for the Credit Cards portfolio experienced a slight decrease, moving from **16.6%** in the 2025 Stress Test Results to **16.3%** in the 2026 Stress Test Results.

This modest improvement is driven by a slightly less severe shock to the U.S. unemployment rate, a key driver of consumer credit performance. While both scenarios feature a peak unemployment rate of 10%, the 2026 Stress Test Scenario specifies a **5.5 percentage point** increase from the starting point. This is less severe than the 2025 Stress Test Scenario, which assumed a **5.9 percentage point** increase in the unemployment rate. The less pronounced shock to unemployment in the 2026 scenario directly contributes to the marginal decrease in projected credit card losses.

Commercial Real Estate, Domestic (CRE) Portfolio

In contrast, Citigroup's projected loan loss rate for the domestic CRE portfolio increased significantly, rising from **8.3%** in the 2025 Stress Test Results to **10.9%** in the 2026 Stress Test Results.

This substantial increase in projected losses is a direct result of the significantly more stressful assumptions for the commercial real estate market in the 2026 scenario. The 2026 Stress Test Scenario mandates a severe **39% decline** in commercial real estate prices. This is a markedly more severe shock compared to the 2025 Stress Test Scenario, which assumed a **30% decline** in CRE prices. The intensified stress on

CRE valuations is the primary driver of the higher projected loan losses for this portfolio year-over-year.

[Usage] Prompt Tokens: 168,376 | Output Tokens: 2,336
[Cost] Input Cost: \$0.210470 | Output Cost: \$0.023360 | Total Cost: \$0.233830
→ Macro numbers found. Running DeepEval Economic Reasoning Check...
→ DeepEval Score: 70.0% | Reason: The response accurately connects macroeconomic severity changes to loss rate changes for both portfolios, with coherent directional logic: a less severe unemployment shock (5.5pp vs 5.9pp) explains the slight credit card loss decrease (16.6% to 16.3%), and a more severe CRE price decline (39% vs 30%) explains the CRE loss increase (8.3% to 10.9%). The economic domain knowledge (unemployment for consumer credit, CRE prices for commercial real estate) is appropriately applied. However, the response omits the required Markdown table with the exact columns specified in the Input, which is a notable structural omission. Additionally, it relies on domain-inferred drivers without confirming whether the Fed documents explicitly stated them, and lacks explicit YoY percentage point change calculations formatted as required (e.g., -0.3pp and +2.6pp). These omissions reduce completeness relative to the Input requirements.
→ DeepEval Usage: Input 2,399 / Output 567 | Total Cost: \$0.026170
✅ Validation: PASSED

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🏃 RUNNING COMPLEX_BRIEF PROMPT | Model: claude-sonnet-4-6

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[Output]
Executive Summary: Citigroup Loan Loss Rate Variance Analysis
Credit Cards & Commercial Real Estate – 2025 vs. 2026 CCAR Stress Test Results

I. Summary of Findings

Citigroup's stressed loan loss rates shifted materially between the 2025 and 2026 CCAR cycles across two key portfolios. The **credit card loss rate was essentially flat** (16.6% in 2025 vs. 16.3% in 2026), while the **commercial real estate (CRE) domestic loss rate increased significantly** (8.3% in 2025 vs. 10.9% in 2026). These movements are directly attributable to specific, quantifiable changes in the macroeconomic severity parameters embedded in the respective severely adverse scenarios, as well as to growth in Citigroup's underlying loan balances.

II. Citigroup Specific Loss Rate Data (Source: Appendix A, Both CCAR Results Documents)

| Portfolio | 2025 CCAR Loss Rate | 2026 CCAR Loss Rate | Year-over-Year Change |

	----	----	----	----
	Credit Cards	**16.6%**	**16.3%**	**▼ 0.3 pp**
	Commercial Real Estate, Domestic	**8.3%**	**10.9%**	**▲ 2.6 pp**

Source: Table A.8 (Citigroup), 2025 CCAR Results; Table A.9 (Citigroup), 2026 CCAR Results.

Additionally, Citigroup's absolute dollar loan losses in these categories changed as follows:

	Portfolio	2025 CCAR Losses (\$B)	2026 CCAR Losses (\$B)	Year-over-Year Change
	----	----	----	----
	Credit Cards	**\$29.6B**	**\$29.7B**	**▲ \$0.1B**
	Commercial Real Estate, Domestic	**\$2.1B**	**\$2.6B**	**▲ \$0.5B**

The near-flat credit card dollar loss despite a marginally lower loss rate reflects modest growth in Citigroup's credit card portfolio balances. The CRE dollar loss increase reflects both a higher loss rate and modest balance growth.

III. Macroeconomic Scenario Severity: Key Variable Comparison

The following table isolates the scenario variables most directly relevant to credit card and CRE loss performance, drawn from the published severely adverse scenario tables in both Stress Test Scenarios documents.

A. Variables Driving Credit Card Loss Rates (Unemployment & Consumer Stress)

	Scenario Variable	2025 Severely Adverse	2026 Severely Adverse	Change
	----	----	----	----
	Unemployment Rate – Peak Level	**10.0%**	**10.0%**	No change
	Unemployment Rate – Jump-off (Q4 prior year)	**4.1%** (Q4 2024)	**4.5%** (Q4 2025)	**▲ 0.4 pp higher starting point**
	Unemployment Rate – Peak Increase (pp)	**↑ 5.9 pp**	**↑ 5.5 pp**	**▼ 0.4 pp less severe**
	Real GDP – Peak-to-Trough Decline	**↓ 7.8%**	**↓ 4.6%**	**▼ 3.2 pp less severe**
	Equity Prices – Decline	**↓ 50%**	**↓ 58%**	**▲ 8 pp more severe**
	CPI Inflation – Trough (annualized)	**~1.3%** (Q3 2025)	**~1.0%** (Q4 2026)	Marginally lower

Sources: Table 2 (Key Variables Comparison), 2025 CCAR Results; Table 2 (Key Variables Comparison), 2026 CCAR Results; Table 4.A, 2025 Stress Test Scenarios; Table 4.A, 2026 Stress Test Scenarios.

B. Variables Driving CRE Loss Rates (Real Estate & Credit Market Stress)

Scenario Variable	2025 Severely Adverse Change	2026 Severely Adverse Change
Commercial Real Estate Prices – Trough Decline	**↓ 30%**	**↓ 39%**
	▲ 9 pp MORE severe	
House Prices – Trough Decline	**↓ 33%**	**↓ 30%**
		▼ 3 pp less severe
BBB Corporate Bond Spread – Peak Level	**5.0%**	**5.7%**
	pp more severe**	**▲ 0.7
BBB Corporate Bond Spread – Jump-off-to-Peak Increase	**↑ 3.9 pp**	
	↑ 4.7 pp	**▲ 0.8 pp more severe**
3-Month Treasury Rate – Trough	**0.1%**	**0.1%**
		No change
10-Year Treasury Yield – Trough	**1.0%**	**2.3%**
		▲ 1.3 pp HIGHER (less rate relief)
10-Year Treasury Yield – Jump-off-to-Trough Decline	**↓ 3.3 pp**	
	↓ 1.8 pp	**▼ 1.5 pp smaller decline**

Sources: Table 2 (Key Variables Comparison), 2025 CCAR Results; Table 2 (Key Variables Comparison), 2026 CCAR Results; Table 4.A, 2025 Stress Test Scenarios; Table 4.A, 2026 Stress Test Scenarios.

IV. Attribution Analysis

A. Credit Cards: Loss Rate Essentially Flat (16.6% → 16.3%, ▼ 0.3 pp)

The near-stability of Citigroup's credit card loss rate across the two cycles is explained by **offsetting macroeconomic forces** that largely cancelled each other out.

Factors that would have reduced credit card losses (favorable):

1. **Smaller unemployment rate increase:** The 2026 severely adverse scenario features a 5.5 percentage point rise in unemployment (from 4.5% to 10.0%), compared to a 5.9 percentage point rise in 2025 (from 4.1% to 10.0%). Credit card losses are highly sensitive to unemployment, as consumer ability to service revolving debt deteriorates sharply with job losses. A smaller incremental shock to unemployment, all else equal, would reduce expected default rates on consumer credit.

2. **Shallower real GDP contraction:** The 2026 scenario features a peak-to-trough real GDP decline of only 4.6%, compared to 7.8% in the 2025 scenario – a 3.2 percentage point reduction in economic severity. A less severe recession implies less aggregate income destruction and, consequently, less consumer credit stress.

Factors that would have increased credit card losses (adverse):

1. **Higher jump-off unemployment rate:** The 2026 scenario begins from a higher unemployment base of 4.5% (Q4 2025) versus 4.1% (Q4 2024). While

the peak level is identical at 10%, a higher starting unemployment rate implies that consumers were already under greater labor market pressure entering the stress horizon, which could sustain elevated delinquency rates throughout the projection period.

2. **Larger equity price decline:** The 2026 scenario features a 58% decline in equity prices versus 50% in 2025. While equity prices are more directly relevant to wealth effects and trading book losses, a sharper equity market collapse can amplify consumer confidence deterioration and reduce the ability of higher-income cardholders to service balances.

3. **Wider BBB corporate bond spreads:** The peak BBB spread of 5.7% in 2026 versus 5.0% in 2025 signals more severe corporate credit stress, which can feed through to consumer employment conditions and indirectly pressure card performance.

Net result: The favorable factors (smaller unemployment increase, shallower GDP contraction) approximately offset the adverse factors (higher starting unemployment, larger equity decline, wider credit spreads), producing a marginally lower credit card loss rate of 16.3% in 2026 versus 16.6% in 2025. The 2026 CCAR Results document explicitly confirms this dynamic, noting that "greater loan losses" in 2026 were driven by "loan balance growth" and "more severe paths for certain scenario variables, including more severe changes in corporate bond spreads," while simultaneously acknowledging that the unemployment rate increase was smaller. For credit cards specifically, these forces balanced, leaving Citigroup's rate nearly unchanged.

B. Commercial Real Estate, Domestic: Loss Rate Increased Materially (8.3% → 10.9%, ▲ 2.6 pp)

The 31% increase in Citigroup's domestic CRE loss rate is directly and primarily attributable to a **substantially more severe CRE price shock** in the 2026 scenario, compounded by a **less favorable interest rate environment** that reduces the refinancing relief available to distressed CRE borrowers.

Primary driver – CRE price shock severity:

The single most important variable for CRE loan loss rates is the magnitude of the decline in commercial real estate prices, which directly determines collateral values and loss-given-default for secured CRE loans. The 2026 severely adverse scenario specifies a **39% decline in CRE prices** from the Q4 2025 jump-off, compared to only a **30% decline** in the 2025 severely adverse scenario from Q4 2024. This represents a **9 percentage point increase in CRE price severity** – the largest directional change of any single scenario variable between the two cycles.

The 2026 Stress Test Scenarios document explicitly explains the rationale for this more severe CRE shock: the scenario was calibrated "near or in the upper one-third" of the severity ranges specified in the Board's

Proposed 2025 Scenario Design Policy Statement, with the CRE trough decline of 39% described as "near the top one-third of the range of 30 to 45 percent." The 2026 CCAR Results document further confirms that "more severe changes in... commercial real estate prices" were a primary driver of increased loan losses year-over-year.

When CRE prices fall 39% rather than 30%, the loan-to-value ratios on Citigroup's CRE portfolio deteriorate far more severely, increasing both the probability of default (as borrowers face negative equity and refinancing barriers) and the loss-given-default (as collateral recovery values are lower). For a bank with Citigroup's international CRE exposure profile – which showed a CRE loss rate of 8.3% in 2025 already reflecting meaningful stress – a 9 percentage point increase in the price shock translates directly into a materially higher loss rate.

****Secondary driver – Less interest rate relief in 2026:****

CRE borrowers benefit from declining interest rates, which reduce refinancing costs and can support property valuations through lower capitalization rates. The 2026 scenario provides ****significantly less interest rate relief**** than the 2025 scenario:

- The ****10-year Treasury yield troughs at 2.3%**** in the 2026 scenario versus ****1.0%**** in the 2025 scenario – a 1.3 percentage point higher floor, meaning long-term rates remain elevated throughout the stress horizon.
- The ****jump-off-to-trough decline in the 10-year Treasury yield is only 1.8 percentage points**** in 2026, compared to ****3.3 percentage points**** in 2025 – a 45% smaller rate relief.

The 2026 CCAR Results document explicitly identifies this as a key driver of results: "Projected capital decreased due to smaller projected unrealized gains on available-for-sale (AFS) securities as a result of smaller projected declines in interest rates in the hypothetical scenario: The 2026 severely adverse scenario had smaller projected declines in interest rates than last year's hypothetical scenario, resulting in elevated interest rate levels throughout the projection horizon."

For CRE specifically, higher sustained interest rates compound the stress from falling property prices. CRE borrowers facing loan maturities during the stress horizon encounter both depressed collateral values ***and*** higher refinancing rates, dramatically increasing default probability. This "double squeeze" – falling asset values combined with elevated financing costs – is a well-established amplifier of CRE credit losses and directly explains why Citigroup's CRE loss rate increased by 2.6 percentage points despite the unemployment shock being marginally smaller.

****Tertiary driver – Wider corporate bond spreads:****

The BBB corporate bond spread peaks at ****5.7%**** in the 2026 scenario versus ****5.0%**** in 2025, an increase of 0.7 percentage points. Wider credit spreads reflect greater perceived credit risk across the economy

and translate into higher borrowing costs for CRE borrowers seeking to refinance or recapitalize distressed properties. This further reduces the ability of CRE borrowers to cure defaults and increases loss severity.

V. Aggregate Context: System-Wide CRE Loss Trends Corroborate Citigroup-Specific Findings

The Citigroup-specific findings are consistent with system-wide trends reported in the aggregate results. Across all 32 banks in the 2026 stress test, the **aggregate CRE domestic loss rate increased from 7.2% (2025, 22 banks) to 8.8% (2026, 32 banks)**. While the sample composition differs (22 banks in 2025 vs. 32 banks in 2026, limiting direct comparability), the directional increase in CRE loss rates is consistent across the system, confirming that the more severe CRE price shock in the 2026 scenario was the dominant driver of higher CRE losses industry-wide. Citigroup's 2.6 percentage point increase (8.3% to 10.9%) exceeds the aggregate increase, suggesting Citigroup's CRE portfolio has above-average sensitivity to CRE price declines, consistent with its international and urban-concentrated CRE exposure profile.

VI. Conclusion

Driver	Credit Cards	CRE Domestic
CRE price shock (↓30% → ↓39%)	Minimal direct impact	**Primary driver of loss rate increase**
Unemployment increase (5.9 pp → 5.5 pp)	**Favorable offset** (smaller shock)	Modest favorable offset
Real GDP contraction (↓7.8% → ↓4.6%)	**Favorable offset** (less severe)	Modest favorable offset
10-year Treasury trough (1.0% → 2.3%)	Minimal direct impact	**Secondary adverse driver** (less rate relief)
BBB spread peak (5.0% → 5.7%)	Modest adverse factor	**Tertiary adverse driver**
Equity price decline (↓50% → ↓58%)	Modest adverse factor	Minimal direct impact
Net YoY Change in Loss Rate	**▼ 0.3 pp (flat)**	**▲ 2.6 pp (material increase)**

The 2026 CCAR results for Citigroup reflect a scenario architecture that deliberately intensified CRE-specific stress – calibrating the CRE price decline near the top of the Board's proposed severity range at 39% – while simultaneously providing less interest rate relief through a higher long-term rate floor. These two forces combined to drive Citigroup's domestic CRE loss rate from 8.3% to 10.9%. Credit card losses, by contrast, were governed primarily by the unemployment trajectory, where the smaller incremental shock (5.5 pp vs. 5.9 pp) and shallower GDP contraction offset the adverse effects of a higher starting unemployment rate and wider credit spreads, leaving the loss rate essentially unchanged at 16.3%.

[Usage] Prompt Tokens: 164,900 | Output Tokens: 3,912

[Cost] Input Cost: \$0.494700 | Output Cost: \$0.058680 | Total Cost: \$0.553380

→ Macro numbers found. Running DeepEval Economic Reasoning Check...

→ DeepEval Score: 90.0% | Reason: The response strongly aligns with the evaluation steps. It accurately identifies the changes in macroeconomic severity (CRE price decline worsening from -30% to -39%, unemployment shock easing from 5.9pp to 5.5pp, GDP contraction shallowing from -7.8% to -4.6%, higher 10-year Treasury trough) and links them coherently to the loss rate changes (credit cards flat at 16.6%→16.3%, CRE rising 8.3%→10.9%). The direction and magnitude of relationships are logical: the flat credit card rate is explained via offsetting forces, and the material CRE increase is tied primarily to the more severe CRE price shock plus reduced rate relief. The economic domain knowledge (LTV deterioration, collateral value effects, refinancing squeeze, unemployment sensitivity of revolving debt) is valid and consistent with the portfolios. The output avoids clear contradictions and even corroborates findings with system-wide trends. Minor shortcomings: some specific numeric values and quotes are asserted as directly from documents and cannot be fully verified here, and the 'international CRE exposure' characterization introduces a mildly unsupported inference, slightly weakening the constraint against unsupported claims.

→ DeepEval Usage: Input 7,070 / Output 668 | Total Cost: \$0.052050

✅ Validation: PASSED